THE ETALE LOCUS IS COMPATIBLE WITH BASE CHANGE UNDER FLATNESS OR FINITE PRESENTATION HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The unramified locus is compatible with base change. Lemma \ref{lemma-set-points-where-fibres-unramified}, cited at proof lines 8434-8440. Gives the equality of the two unramified point sets.
- A morphism is etale at a point exactly when it is flat and unramified there. Lemma \ref{lemma-flat-unramified-etale}, cited at proof lines 8443-8444. Turns the unramified statement into the etale statement in case (1).
- Flatness is preserved by base change and by composition, and is detected at the source point. Lemma \ref{lemma-base-change-module-flat}, Lemma \ref{lemma-composition-module-flat} and Lemma \ref{lemma-flat-permanence}, cited at proof lines 8447-8453. Transports flatness in case (2).
- The etale-at-a-point criterion. Lemma \ref{lemma-etale-at-point}, cited at proof lines 8454-8456. Concludes that the morphism is etale at the image point in case (2).

Source reference. Original lemma 0476: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5497–5497. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0475: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5496–5496. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3242–3242. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01U8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2401–2401. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01U6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2399–2399. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3354–3354. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2440–2440. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3241–3241. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian diagram of schemes. Let $W \subset X$, resp. $W' \subset X'$ be the open subscheme of points where f, resp. f' is étale. Then $W' = (g')^{-1}(W)$ if

(1) *f is flat and locally of finite presentation, or*
(2) *f is locally of finite presentation and g is flat.*

Proof. Assume first that f locally of finite type. Consider the set

$$T = \{x \in X \mid f \text{ is unramified at } x\}$$

and the corresponding set $T' \subset X'$ for f'. Then $T' = (g')^{-1}(T)$ by Lemma 0475.

Thus case (1) follows because in case (1) T is the (open) set of points where f is étale by Lemma 02GV.

In case (2) let $x' \in W'$. Then g' is flat at x' (Lemma 01U8) and $g \circ f'$ is flat at x' (Lemma 01U6). It follows that f is flat at $x = g'(x')$ by Lemma 02JZ. On the other hand, since $x' \in T'$ (Lemma 01VB) we see that $x \in T$. Hence f is étale at x by Lemma 02GU. □